

Homework #3

For the questions below, use the following constants:

Speed of light: $c=2.998 \times 10^8$ m/s
Electron charge: $e=1.602 \times 10^{-19}$ C
Plank constant: $h=6.626 \times 10^{-34}$ J/Hz

- 1- An EDFA power amplifier produces output power of $P_{s,out} = 26$ dBm for an input level of 1.8 dBm at 1542 nm. Please find the amplifier gain.
- 2- A DWDM optical transmission system is designed to have 100 GHz channel spacing. How many wavelength channels can be utilized in the 1536 to 1556 nm spectral band?
- 3- Consider an EDFA with maximum 26 dB gain and maximum output power of 0 dBm. Please calculate the actual EDFA gains when 1, 2, 3 or 8 wavelength channels are multiplexed in WDM systems. Each channel input power is 0.001 mW. (Hint: when the output power is lower than the maximum output power, the EDFA is not saturated and the actual gain is the maximum gain; otherwise the actual gain is smaller than the maximum gain)
- 4- Please calculate pulse spread ΔT caused by polarization-mode dispersion (PMD) if $D_{PMD} = 0.2$ ps/ $\sqrt{\text{km}}$ and fiber length is $L = 120$ km. (Hint: ΔT caused by PMD is proportional to $\sqrt{L}$)
- 5- Assume an optical signal is operated at a carrier frequency of 193.5 THz, whose bandwidth is 80-GHz wide.
 - (a) What is the pulse broadening ΔT after 80 km long single mode fiber propagation, with the dispersion coefficient D of 17 ps/nm/km?
 - (b) If $D = -80$ ps/nm/km, what is the pulse broadening ΔT after 20-km propagation?
 - (c) What is the difference between these two cases? (Hint: you could describe in each case, whether longer wavelength or shorter wavelength transmits faster.)